

Experimental Investigation of Bending and Tensile Strengths of 3D Printed Specimens with Varying Infill Geometries

Introduction & Motivation

This experiment tests 3 different infill geometries: grid, honeycomb, and triangular, (Figure 1) manufactured at equal infill densities, to determine which infill is the strongest on a mass basis under tension and bending.

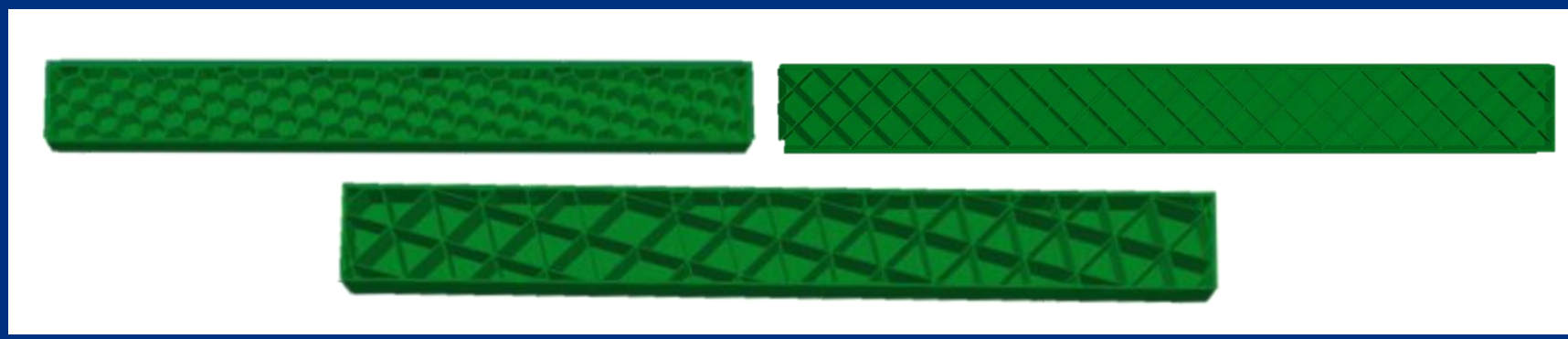


Figure 1. Infill geometries Honeycomb (top left), grid (top right), and triangular (bottom)

Points of Discussion

- Manufacturing-time normalization shows triangular is most efficient for tension, while honeycomb is best for bending despite longer print time.
- Tension test fracture point occurred at interface between infill geometry and clamping region (with 100% infill).
- The prints were manufactured at the Aero Maker Space, and had impurities
- Observations made in the lab confirm our calculated results – Triangular dog bone was able to withstand the most displacement before fracture, etc.

Results

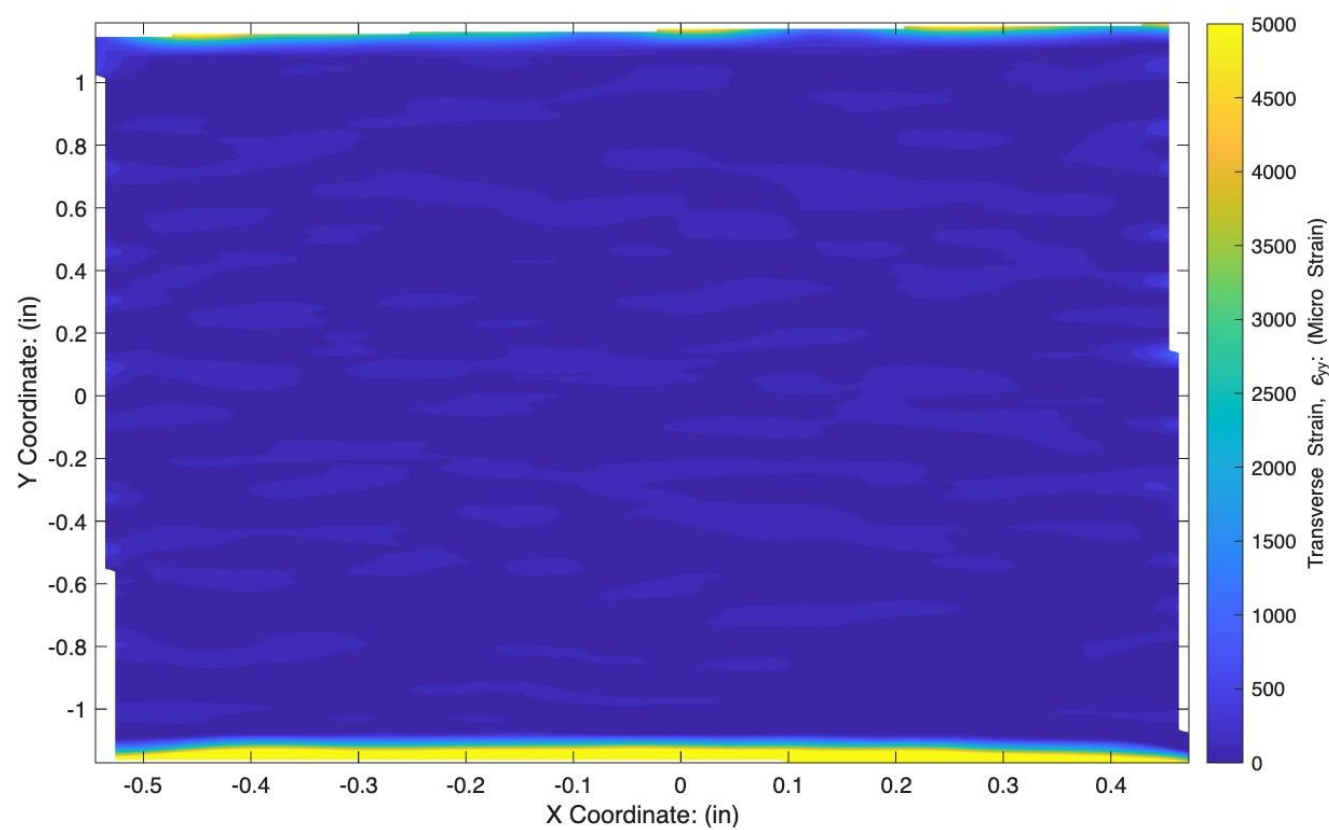


Figure 2. Tensile Specimen Honeycomb-1 ϵ_{yy}

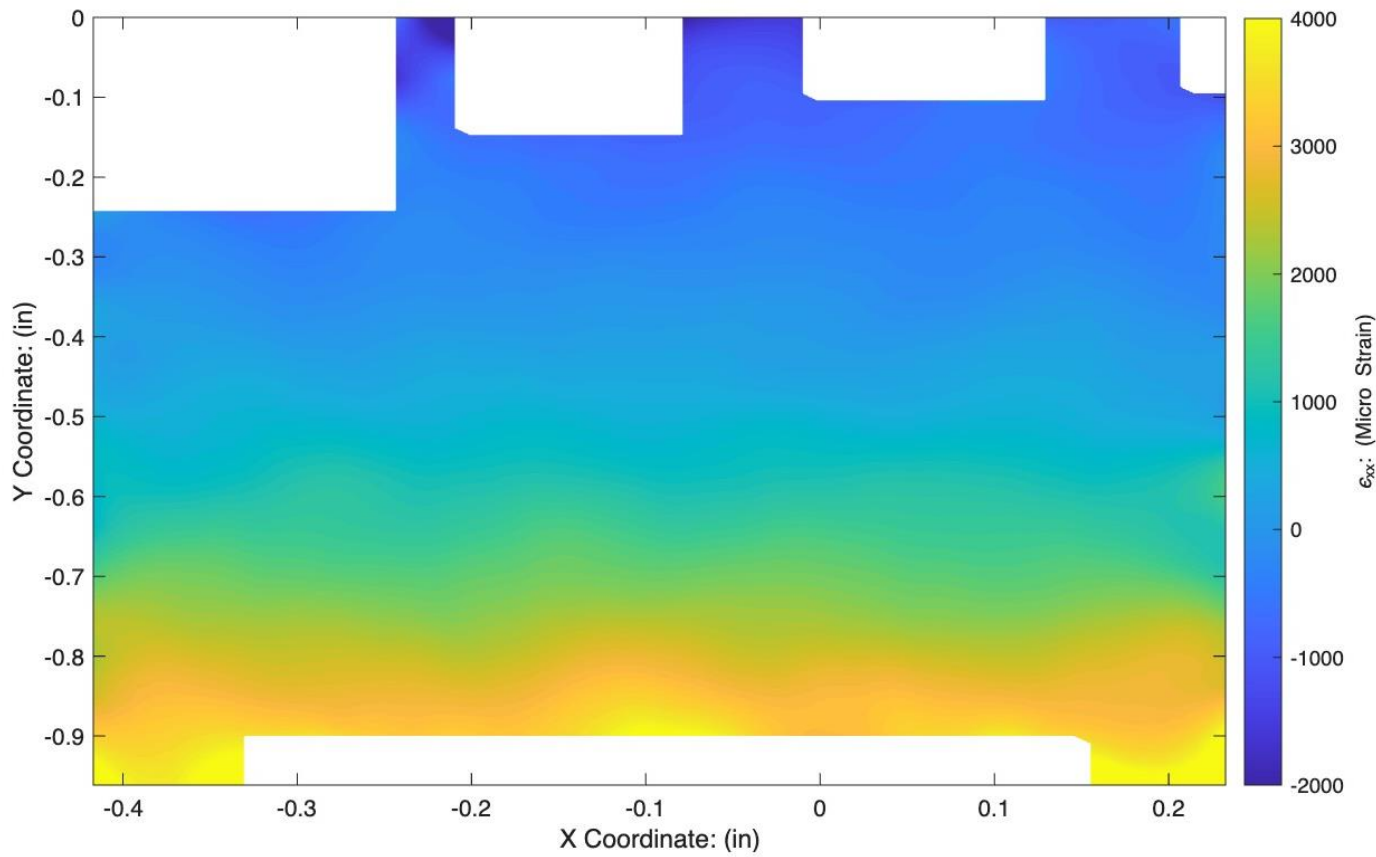


Figure 3. Honeycomb-1 Beam Specimen Axial Micro Strain ϵ_{xx}

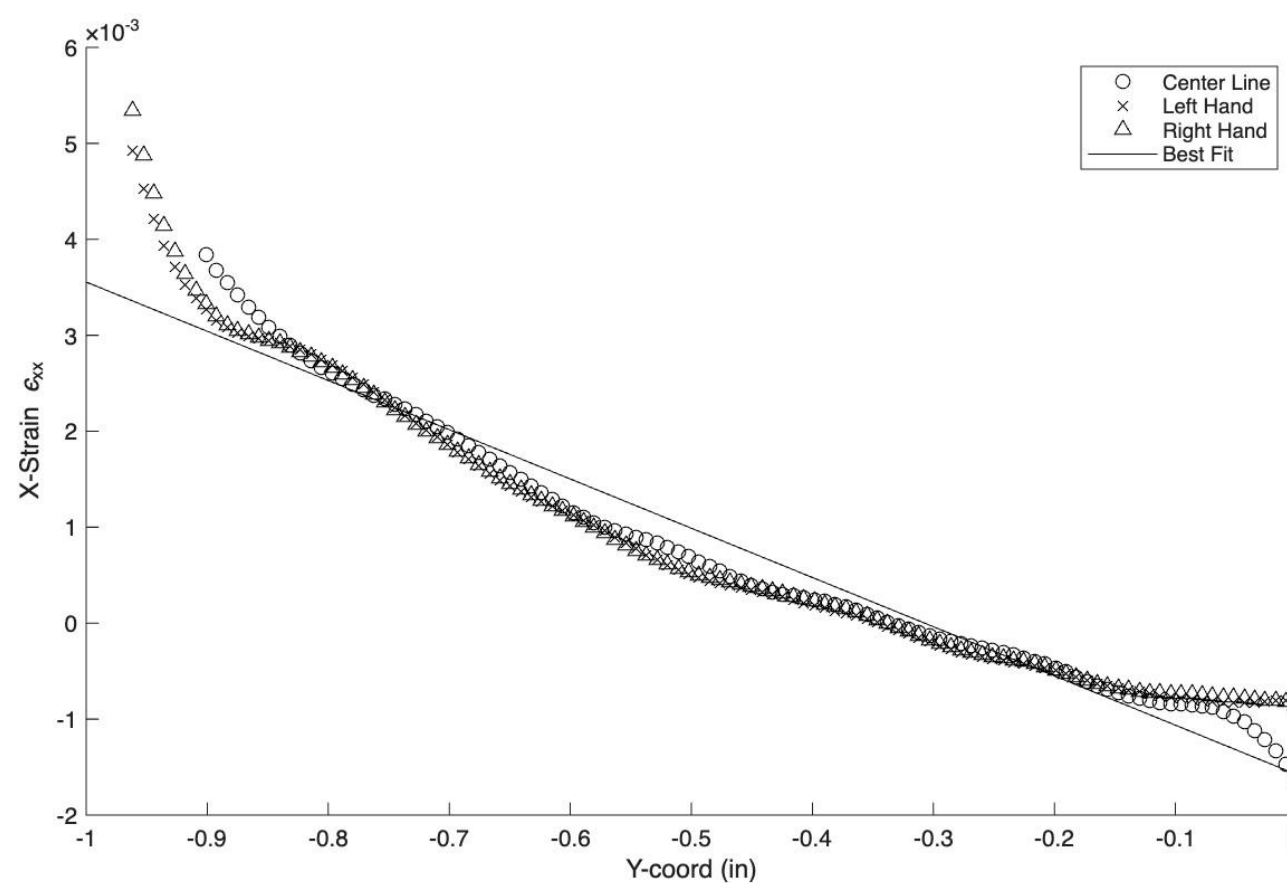


Figure 4. Bending Specimen Honeycomb-1 Axial Strain Data as Function of Y-Coordinate

Geometry	Modulus (MPa)	Normalized Manufacturing Time
Tension		
1. Triangular	$E_T = 2.5812 \times 10^3$	1.01x
2. Grid	$E_T = 2.1773 \times 10^3$	1.00x
3. Honeycomb	$E_T = 1.3784 \times 10^3$	1.18x
Bending		
1. Honeycomb	$E_C = 1.6552 \times 10^5$	1.49x
2. Triangular	$E_C = 1.4126 \times 10^5$	1.07x
3. Grid	$E_C = 1.2899 \times 10^5$	1.00x

Table I. Elastic modulus results from experimentation

Conclusion

- Triangular infill provided the highest tensile stiffness and was the most time-efficient option for tension-dominated loading.
- Honeycomb infill exhibited the greatest flexural stiffness, making it the best choice for bending applications despite its longer print time.
- Grid infill printed fastest but consistently showed the lowest stiffness in both loading conditions.
- Overall, the optimal infill geometry depends on whether tensile or bending performance is most critical for the intended structure.

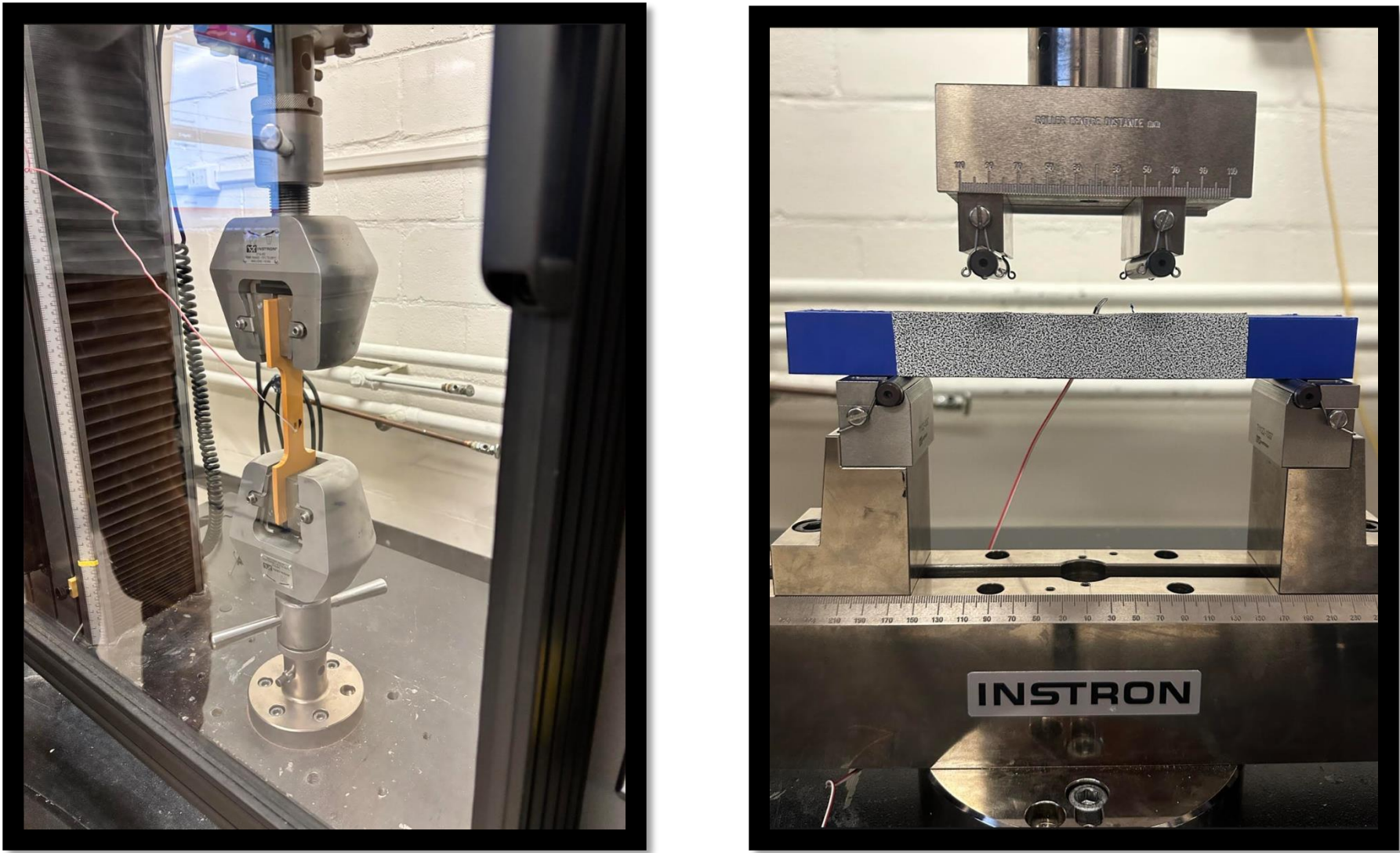


Figure 2. Experimental setup for tensile test (left) and bending test (right)

Experimental Setup

- Tests performed using Instron load frame and DIC system (Figure 2).
- Printed dog bones (tension) and square beams (bending) for all three infill types.
- Ran each test twice, totaling 12 experiments.
- Used strain gauges on first tension tests to identify elastic region and failure point.
- Captured DIC images through the elastic region for both tests.
- Processed data using VIC-2D and MATLAB to compute elastic moduli and compare geometries.